

CS 171: Discussion 2 (Sept 8)

1. Equivalence of Definitions

You are given an encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ that satisfies the condition

$$\Pr[M = m | C = c] = \Pr[M = m]$$

for every probability distribution M over $\mathcal{M}$, every message $m \in \mathcal{M}$, and every ciphertext $c \in \mathcal{C}$ such that $\Pr[C = c] > 0$. Show that for any two messages $m, m' \in \mathcal{M}$ and for any ciphertext $c \in \mathcal{C}$,

$$\Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c].$$

Solution Fix any two messages $m, m' \in \mathcal{M}$ and any ciphertext $c \in \mathcal{C}$. First suppose that $\Pr[C = c] > 0$. By Bayes' rule,

$$\begin{aligned} \Pr[C = c | M = m] &= \frac{\Pr[M = m | C = c] \Pr[C = c]}{\Pr[M = m]} \\ &= \frac{\Pr[M = m] \Pr[C = c]}{\Pr[M = m]} \\ &= \Pr[C = c], \end{aligned}$$

where the second equality follows from the assumed definition of perfect secrecy. Conditioning on $M = m$ means that the fixed message m is encrypted under the random key K , so

$$\Pr[\text{Enc}_K(m) = c] = \Pr[C = c | M = m] = \Pr[C = c].$$

By the exact same argument applied to m' , we obtain

$$\Pr[\text{Enc}_K(m') = c] = \Pr[C = c].$$

Thus, $\Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c]$.

The case $\Pr[C = c] = 0$.

$$0 = \Pr[C = c] = \sum_{\tilde{m} \in \mathcal{M}} \Pr[C = c | M = \tilde{m}] \Pr[M = \tilde{m}].$$

Every term in this sum is nonnegative. Therefore, every term whose message has positive probability must be zero. In particular, since $\Pr[M = m] > 0$ and $\Pr[M = m'] > 0$,

$$\Pr[C = c | M = m] = 0 \quad \text{and} \quad \Pr[C = c | M = m'] = 0.$$

Equivalently,

$$\Pr[\text{Enc}_K(m) = c] = 0 = \Pr[\text{Enc}_K(m') = c].$$

Thus, the desired equality also holds when $\Pr[C = c] = 0$, and hence for every ciphertext c . $\square$

2. Non-Negligible Function

A function $f : \mathbb{Z}^+ \rightarrow [0, 1]$ is a *negligible function* if $\forall$ polynomials $p(\cdot)$, $\exists N \in \mathbb{Z}^+$ such that $\forall n > N$ we have $f(n) < \frac{1}{p(n)}$.

See the section on *negation* at https://en.wikipedia.org/wiki/Universal_quantification.

Define a non-negligible function using the negation of the definition of a negligible function.

Solution A function $f : \mathbb{Z}^+ \rightarrow [0, 1]$ is a *non-negligible function* if $\exists$ polynomials $p(\cdot)$ such that $\forall N \in \mathbb{Z}^+$, $\exists n > N$ such that we have $f(n) \geq \frac{1}{p(n)}$. $\square$

3. A Different One-time Pad

For set S , let $s \xleftarrow{\$} S$ denote selecting a uniformly random element s from S . Consider the following encryption scheme for the message space $\{0, 1\}$.

- **Gen**: Choose two random bits $a, b \xleftarrow{\$} \{0, 1\}$. Output (a, b) .
- **Enc** $((a, b), m)$: Choose random $x_1 \xleftarrow{\$} \{0, 1\}$ and compute x_2 such that $a \cdot x_1 + b + x_2 = m$ where $+$ and $\cdot$ are operations over $\mathbb{Z}_2$, aka the integers mod 2. Output (x_1, x_2) .
- **Dec** $((a, b), (x_1, x_2))$: Output $m = a \cdot x_1 + b + x_2$.

- (a) Show that this scheme is perfectly secure. *Hint: Use the second (equivalent) definition of perfect secrecy from Q1.*
- (b) Recall that the one-time pad you saw in class can be broken under a *known-plaintext attack*: upon receipt of both a plaintext and corresponding ciphertext for a message, the attacker can recover information about the key.
 - (i) Is the scheme above also susceptible to a known-plaintext attack?
 - (ii) For the scheme from class, how many plaintext-ciphertext pairs are necessary to recover the entire key? How about for this new scheme—or is it impossible to fully recover the key, even with unlimited plaintext-ciphertext pairs?

Solution

- (a) Fix any two messages $m, m' \in \{0, 1\}$ and a ciphertext $(x_1, x_2) \in \{0, 1\} \times \{0, 1\}$. We will show that

$$\Pr[\text{Enc}(K, m) = c] = \Pr[\text{Enc}(K, m') = c]$$

Let (A, B, X_1) be the uniform random variables over $\{0, 1\} \times \{0, 1\}$. The random variable denoting the key K is given by (A, B) and the first component of the ciphertext

is X_1 .

$$\begin{aligned}
\Pr[\text{Enc}(K, m) = c] &= \Pr_{K, X_1}(\text{Enc}(K, m) = c) \\
&= \Pr_{A, B, X_1}(AX_1 + B + x_2 = m \wedge X_1 = x_1) \\
&= (1/2) \Pr_{A, B}(Ax_1 + B = m - x_2) \\
&= (1/2) \sum_{a \in \{0,1\}} \Pr_B[ax_1 + B = m - x_2 | A = a] \Pr_A[A = a] \\
&= (1/2) \sum_{a \in \{0,1\}} \Pr_B[B = m - x_2 - ax_1 | A = a] (1/2) \\
&= (1/2)(1/2) = 1/4
\end{aligned}$$

By a similar argument, we can show that $\Pr[\text{Enc}(K, m') = c] = 1/4$.

- (b) (i) Yes. From a known plaintext-ciphertext pair $(m, (x_1, x_2))$, the adversary learns the equation

$$ax_1 + b + x_2 = m,$$

or equivalently $ax_1 + b = m - x_2$, over $\mathbb{Z}_2$. This equation reveals information about the secret key (a, b) :

- If $x_1 = 0$, the equation is $b = m - x_2$, so the adversary learns b .
- If $x_1 = 1$, the equation is $a + b = m - x_2$, so the adversary learns $a + b$.

Thus, the scheme is susceptible to a known-plaintext attack, although one pair alone does not uniquely determine both key bits.

- (ii) For the standard one-time pad, one plaintext-ciphertext pair is enough to recover the entire key: from $c = m \oplus k$, the adversary computes $k = m \oplus c$.

For the new scheme, the adversary needs two linearly independent equations. There are two cases:

- If $x_1 = 0$, then the equation becomes $b + x_2 = m$. Thus, the pair reveals $b = m - x_2$ but gives no information about a . Any additional pair with $x_1 = 0$ gives another equation with the same coefficient vector and still reveals nothing about a .
- If $x_1 = 1$, then the equation becomes $a + b + x_2 = m$. Thus, the pair reveals $a + b = m - x_2$ but does not distinguish the individual values of a and b . Any additional pair with $x_1 = 1$ is again linearly dependent and does not separate them.

Once the adversary has one pair with $x_1 = 0$ and one pair with $x_1 = 1$, it first recovers b from the former and then recovers a from the latter. These are two linearly independent equations in the two unknowns a and b . Therefore, the minimum number of plaintext-ciphertext pairs needed to recover the key is two, but the two pairs must have different values of x_1 . Two pairs do not guarantee recovery: an unlucky adversary may receive the same value of x_1 repeatedly and therefore fail to obtain two linearly independent equations. After observing t pairs, the probability that all observed values of x_1 are identical is 2^{1-t} . Thus, an extremely

unlucky adversary might never see both values, although the probability of still being unable to recover the key approaches 0 as the number of observed pairs grows.

□